

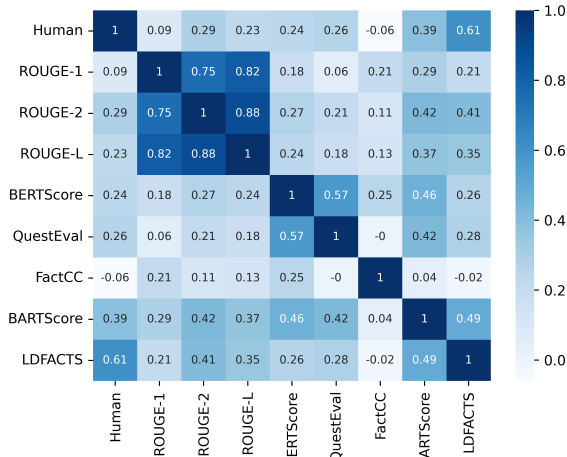


Figure 4: Pairwise Kendall’s tau correlations of metrics for the LongSciEval PubMed data set. LongDocFACTScore is denoted "LDFACTS", human annotations are denoted "Human".

Metric	Time taken (s)
FactCC	24
QuestEval	160
BARTScore	1
LongDocFACTScore	8

Table 5: Time (s) to run each metric on 15 samples.

asures of factual consistency. Interestingly, Figure 4 shows that several automatic metrics have strong correlations with each other, suggesting that there is overlap in what they are measuring, but there is lower correlation between LongDocFACTScore and the other automatic metrics, suggesting that by providing coverage of a long document, LongDocFACTScore captures new information which the other metrics miss.

5.3. Computational Efficiency

In Table 5, we compare the average time taken, in seconds, to run each transformer-based (Vaswani et al., 2017) automatic metric designed to measure factual consistency on fifteen samples from the PubMed LongSciVerify data set. Table 5 shows that LongDocFACTScore, implemented with BARTScore, is second fastest, despite evaluating the generated summary against the entire source document, rather than a truncated version of it. In contrast, QuestEval is shown to be 20x slower, and FactCC 3x slower, than LongDocFACTScore.

5.4. Short Document Data Set Results

Although the intended use of LongDocFACTScore is to evaluate the factual consistency of abstractive

summarisation for long documents, we additionally evaluate LongDocFACTScore against other automatic metrics on a variety of human annotated, short document, abstractive summarisation data sets, to validate its performance in this setting. We repeat the analysis conducted by Yuan et al., (2021) on the data sets containing human measures of factuality, and use their human annotated data and code¹¹, to report the Spearman correlation results for the SummEval data set’s factuality measure (Fabbri et al., 2021), the accuracy scores for the Rank19 data set’s factuality annotations (Falke et al., 2019), and the Pearson correlation between the automatic metrics and the human factuality annotations for the two QAGS data sets (Wang et al., 2020). We used same the measures of correlation for each data set as in the original analysis conducted by Yuan et al., (2021), rather than Kendall’s tau correlations, to enable a direct comparison to their reported scores.

Table 6 gives the results of this analysis. The top section of Table 6 gives the results reported by Yuan et al., (2021), in middle section we report our results, and in the bottom section we re-report G-EVAL (Liu et al., 2023) results, a SOTA metric which uses GPT-4 to calculate factuality scores. In Table 6, it can be seen that LongDocFACTScore performs comparably in its ability to measure factual consistency to the original BARTScore model for short document summaries, indicating that the framework can be used on documents of differing lengths. G-EVAL-4 reports higher correlations results on the SummEval and QAGS-XSUM data sets but slightly lower correlations on QAGS-CNN. They do not report results for the Rank19 data set.

Although GPT-4 has a much greater token limit than standard PLMs, there is still ultimately a limit, therefore, in future work, LongDocFACTScore could be used to extend this metric, or other LLM-based metrics to very long documents, or multi-document settings. Although the token limit is longer for LLMs, non-relevant content can distract LLM-based metrics and degrade performance (Min et al., 2023), therefore we hypothesise that applying the LongDocFACTScore framework, which considers smaller sections of text at one time, could improve the performance of these models. Future work which extends LLM-based metrics should also consider in its analysis the computational cost of running an LLM-based metric in comparison to running more efficient metrics.

5.5. Parameter Study

We study effects that different parameter settings have on the LongDocFACTScore metric. We report the impact of different parameter settings on

¹¹<https://github.com/neulab/BARTScore>

	SE	R19	QAGS	QAGS
	Fact	Acc	CNN	XSum
ROUGE-1	0.16	0.57	0.34	-0.01
ROUGE-2	0.19	0.63	0.46	0.10
ROUGE-L	0.12	0.59	0.36	0.02
MoverScore	0.16	<u>0.71</u>	0.41	0.05
BERTScore	0.11	<u>0.71</u>	0.58	0.02
FactCC	-	0.70	-	-
QAGS	-	0.72	0.55	<u>0.18</u>
BARTScore	0.31	0.68	0.66	0.01
LDFACTS (BARTScore)	<u>0.36</u>	0.68	<u>0.65</u>	0.04
G-EVAL-4	0.51	-	0.63	0.56

Table 6: Correlation between human measures of factuality on short document data sets, including re-reported results (Yuan et al., 2021; Liu et al., 2023). LDFACTS denotes LongDocFACTScore.

LongDocFACTScore setting	Score
BARTScore	0.440
LongDocFACTScore $K = 1$	0.605
LongDocFACTScore $K = 3$	0.610
LongDocFACTScore $K = 5$	0.600
LongDocFACTScore $K = 7$	0.600
LongDocFACTScore $K = 9$	0.595
LongDocFACTScore $K = 11$	0.590
LongDocFACTScore $K = I$	0.575

Table 7: The effect of varying K , the number of similar sentences considered for the LongDocFACTScore calculation, on the Kendall’s tau correlation with human judgements of factuality.

the Kendall’s tau correlations when evaluating the LongSciVerify data set. The PubMed and ArXiv articles are combined for this parameter study.

Table 7 shows the effect of varying K in the LongDocFACTScore framework (i.e., the maximum number of candidate similar source document sentences considered per summary sentence) on the Kendall’s tau correlation with the human measures of factual consistency. The last row, $K = I$, gives the Kendall’s tau correlation when all sentences in the source document are considered. The correlation of BARTScore with human annotations of factual consistency is also provided for reference. $K = 3$ is shown to be the best parameter, however, the effects of varying K are seen to be small. This is somewhat expected as the maximum score of the K text snippets is carried forward in the LongDocFACTScore metric, and it is likely that the highest scoring sentences correlate well with the most similar sentence embeddings.

Although varying K is not shown make a large

LongDocFACTScore setting	Time taken (s)
$K = 3$	8
$K = I$	134
$K = I$ (no similarity calculations)	125

Table 8: Time taken (s) to run LongDocFACTScore on 15 samples, when implemented with different settings.

Method	Score
$s_k^* = s_k$	0.605
$s_k^* = s_{k-1} + s_k + s_{k+1}$	0.610
$s_k^* = s_{k-2} + s_k + s_{k+2}$	0.595

Table 9: The effect of varying the number of source document sentences concatenated for the LongDocFACTScore calculation on the Kendall’s tau correlation with human judgements of factuality.

difference to the performance of the metric, by selecting $K = 3$ candidate sentences, rather than cycling through all sentences in the source document (i.e., $K = I$), the score calculation in LongDocFACTScore is only calculated for approximately 1-2% of sentences from the source articles in the PubMed and ArXiv data sets. Therefore, by increasing the number of candidate similar sentences K , LongDocFACTScore becomes decreasingly efficient and, by extension, less suitable for use on long documents. To illustrate this point, in Table 8 we give the results of the repeated efficiency calculation from Table 5, where LongDocFACTScore is implemented with $K = 3$ and $K = I$. If $K = I$, there is no need to calculate sentence embeddings or perform the sentence similarity calculation, therefore we additionally report the time taken without these calculations. Table 8 shows that, for the LongSciVerify PubMed long document data set, performing the sentence similarity calculation to select the $K = 3$ most similar text snippets speeds up the metric over 15x.

In Table 9, the number of candidate sentences is kept constant at $K = 3$ and the effect of concatenating the source sentence with the previous and following sentence(s) to generate a text snippet is examined on the documents from the LongSciVerify data set. Table 9 shows that although concatenating one sentence either side of a selected sentence performs best, there is little variation in the Kendall’s tau correlation between the different settings.

6. Conclusion

The prevalence of LLMs and other neural methods for abstractive summarisation of long documents in real world settings is rapidly increasing, however, the abstractive methods used to generate these summaries have known issues with factual inconsistency and hallucination. In this work, we begin to address the lack of research into the suitability of automatic evaluation metrics for assessing factual consistency of long document summarisation, and make the following contributions: (i) we show that existing automatic metrics for assessing factual consistency, which have previously shown good performance on short document data sets, do not perform well in long document settings, (ii) we propose a new framework, LongDocFACTScore, which is able to consider an entire source document in its calculation, without the need to truncate it, and outperforms existing SOTA metrics in its ability to correlate with human measures of factual consistency on long document summarisation data sets, whilst still being more efficient than many SOTA automatic evaluation metrics, (iii) we work to address the lack of resources for evaluating automatic metrics in long document settings and release our LongSciVerify data set, designed for evaluating factuality metrics on the long document summarisation task. We hope that this work promotes further research into automatic metrics for evaluating abstractive summarisation of long documents. In future work, we hope to apply the LongDocFACTScore framework to extend other automatic metrics, such as newer LLM-based metrics. We also hope to incorporate our work into wider LLM evaluation frameworks.

Limitations

Firstly, we review the limitations of our human evaluation study. In our study, we recruited expert annotators, as the long document data sets are domain specific. It is difficult to recruit large numbers of expert annotators and therefore an improvement on this work would be to conduct a larger human evaluation study with more annotators evaluating more documents. We also note that two out of three annotators of the ArXiv data set have a first language which is not English, although they are both fluent in English. Furthermore, although the annotators of the ArXiv data set had all achieved a minimum of an undergraduate degree in a physical science, they did not necessarily study physics, which was the domain of most articles randomly sampled for human evaluation.

Secondly, we comment on the limitations of the LongDocFACTScore metric. One issue with this metric, and other SOTA factuality metrics, is that

they favour extractive summaries. Therefore, although this metric is shown to be effective at measuring the factual consistency of long document abstractive summaries, we suggest that this metric is used in conjunction with other metrics, within a wider framework, such as FLASK (Ye et al., 2023). We also note that this evaluation only included English-language data sets.

Lastly, we discuss the computational cost of our work. We were able to monitor our GPU usage and found that for all experiments run in this period, we used approximately 1200 GPU hours. Despite our metric, LongDocFACTScore, being comparably efficient (see Table 5 and Table 8), we acknowledge that working with large neural models, as well as having environmental implications, is not economically possible for many researchers.

Ethics Statement

Throughout our research, we complied with our institution’s ethical guidelines. We used open-source data and software, for which no ethical approvals were required.

In our study, we conduct a human evaluation. As detailed in Section 4, we were fortunate enough to be able to recruit colleagues, who are domain-experts in the field of the data sets. They volunteered to participate in the study without payment, so we did not need to consider the ethics of crowd-worker payment.

Our work proposes a metric for assessing the factual consistency of abstractive summaries generated for long documents. This metric can be used to help researchers assess the performance of their summarisation methods, however, to minimize any harm which may be caused by deploying an abstractive summarisation model in a live setting, we suggest that any method should be thoroughly evaluated by humans in the setting it is intended to be deployed.

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7. Bibliographical References

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